

PROJECT	Credit Card Approval Prediction	TEAM	AI/ML Program
DOCUMENT	Proposed Solution	DATE	July 2026

Proposed Solution

Overview

The proposed solution automates the credit card approval decision using machine learning. Historical applicant data is used to train multiple classification models, the best-performing model is selected, and it is integrated into a Flask web application for real-time predictions.

Solution Components

- Data Collection: Kaggle dataset with applicant records and credit history records
- Data Preprocessing: cleaning, merging, feature engineering, and encoding
- Model Training: four classification algorithms trained and compared
- Model Selection: best model (Random Forest) saved for deployment
- Web Application: Flask-based interface for real-time prediction
- Optional Cloud Deployment: IBM Watson Machine Learning integration

Novelty

Rather than relying on a single algorithm, this solution systematically compares four different classification approaches (a linear model, a single tree, an ensemble of trees, and a gradient-boosted ensemble) and selects the one that best balances precision and recall for the harder-to-predict "rejected" class, given the natural class imbalance in credit approval data.